



We are on a mission to address the digital skills gap for 10 Million+ young professionals, train and empower them to forge a career path into future tech

Auto IT & Methods

APRIL, 2026



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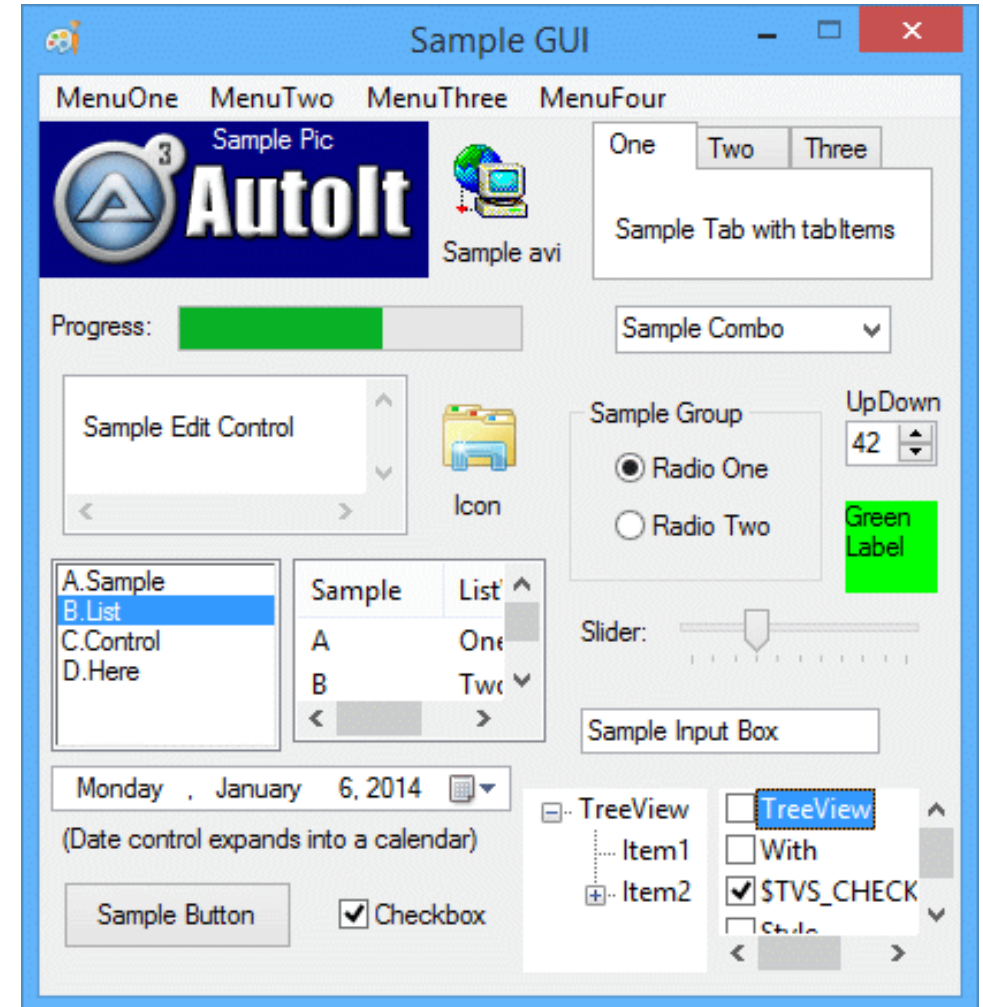
What is Auto IT?



What is Auto IT?

Auto IT

- Auto It is a freeware scripting language designed for automating windows GUI and general scripting.
- It uses a combination of mouse movement, keystrokes and window control manipulation to automate a task which is not possible by selenium web driver.
- Selenium is an open-source tool that is designed to automate web-based applications on different browsers but to handle window GUI and non-HTML popups in application.
- Autolt in Selenium is required as these window-based activity are not handled by Selenium.



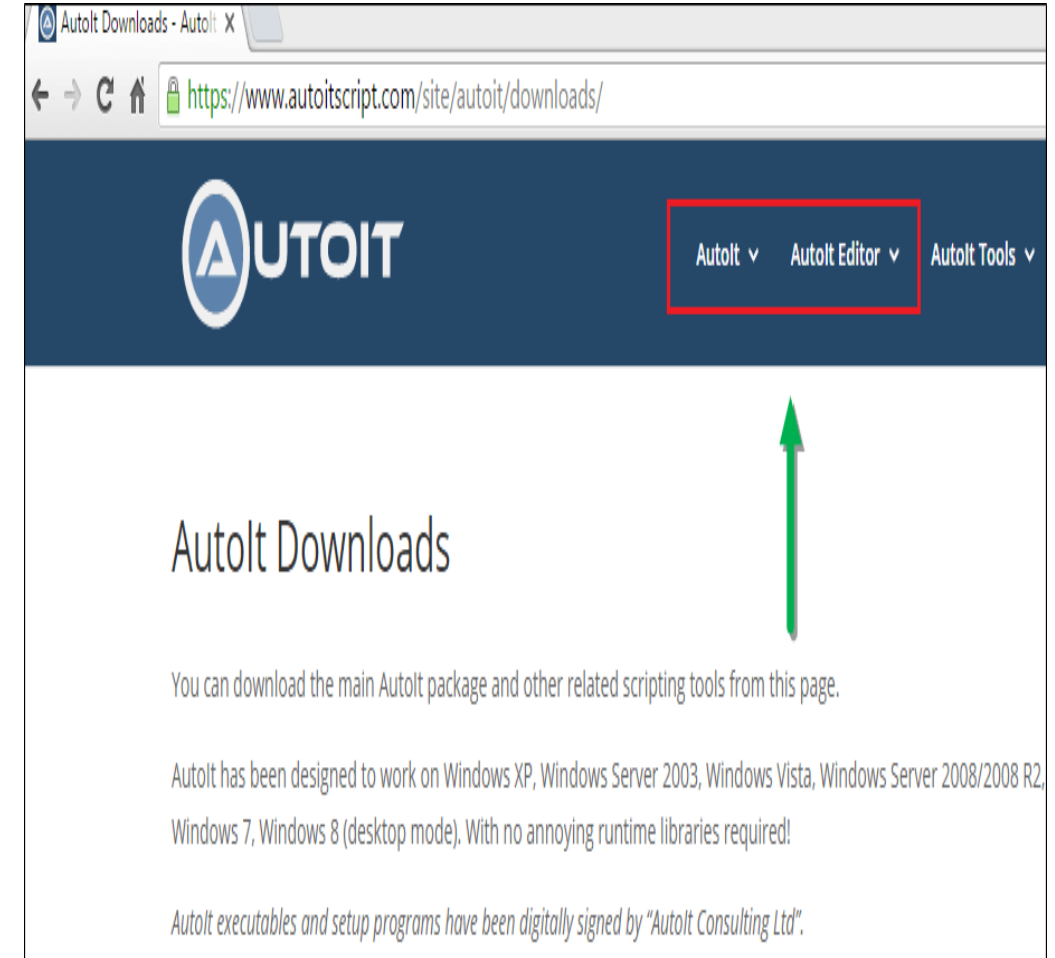
Auto IT Installation



Auto IT Installation

How to download & Install AutoIT?

- Go to the URL –
<https://www.autoitscript.com/site/autoit/downloads/>
- Hover on 'Autoit' and 'Autoit' Editor dropdown.

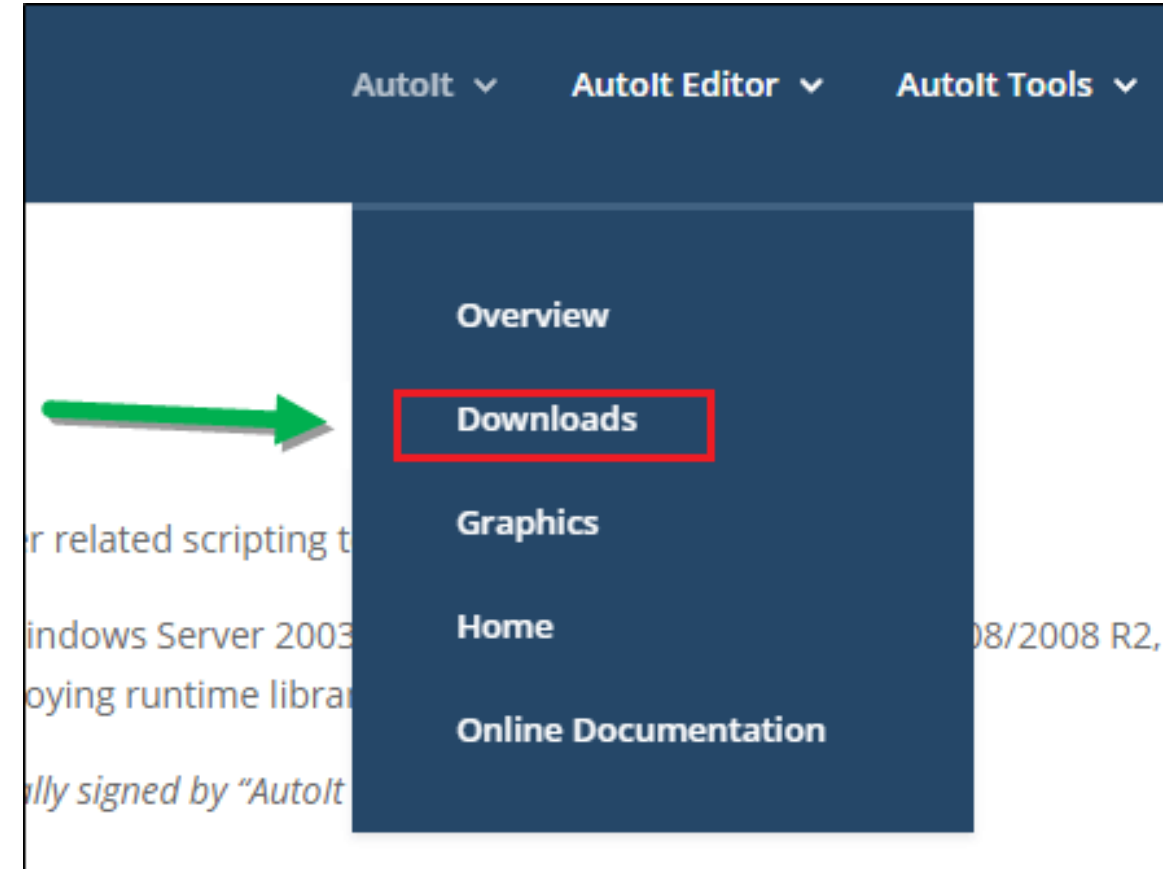


Auto IT Installation

How to download & Install AutoIT? (Contd.)

- Click on 'AutoIT' Downloads option.
- Download "Autoit" by clicking on 'Download Autoit' button .

Software	Download
AutoIt Full Installation. Includes x86 and x64 components, and: • AutoIt program files, documentation and examples. • Aut2Exe – Script to executable converter. Convert your scripts into standalone .exe files! • AutoItX – DLL/COM control. Add AutoIt features to your favorite programming and scripting languages! Also features a C# assembly and PowerShell CmdLets. • Editor – A cut down version of the SciTE script editor package to get started. Download the package below for the full version!	



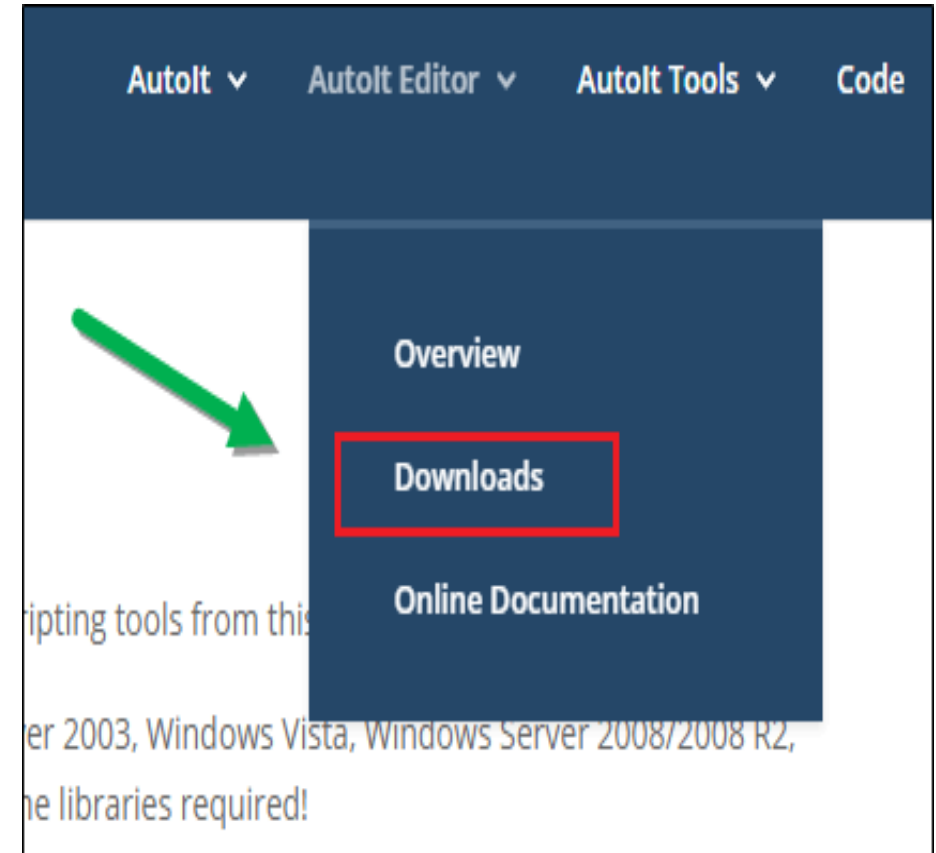
Auto IT Installation

How to download & Install AutoIT? (Contd.)

- Now download "Autoit editor" by clicking on 'Downloads' button .
- Click on the link as shown below.

Current Versions

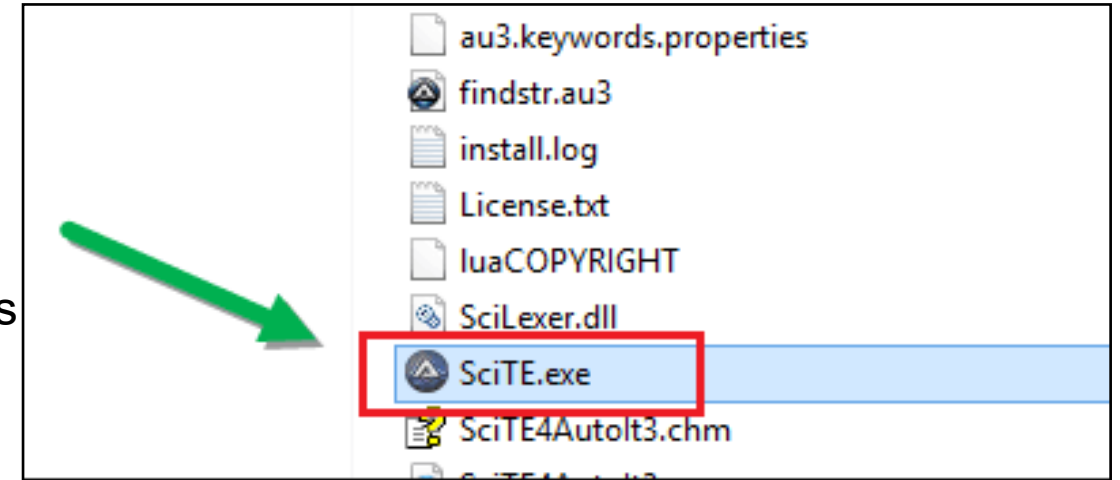
File	Date updated	Notes
SciTE4AutoIt3.exe (5439Kb)	16-3-2021	Installer containing SciTE and all configuration files plus utilities. Update History . Definition files included: AutoIt v3.3.14.5 and BETA v3.3.15.3



Auto IT Installation

How to download & Install AutoIT? (Contd.)

- After successful installation open up AutoIT.
- Go to 'C:\Program Files (x86)\AutoIt3\SciTE'
- Click on 'SciTE.exe' file, the AutoIT editor opens as shown



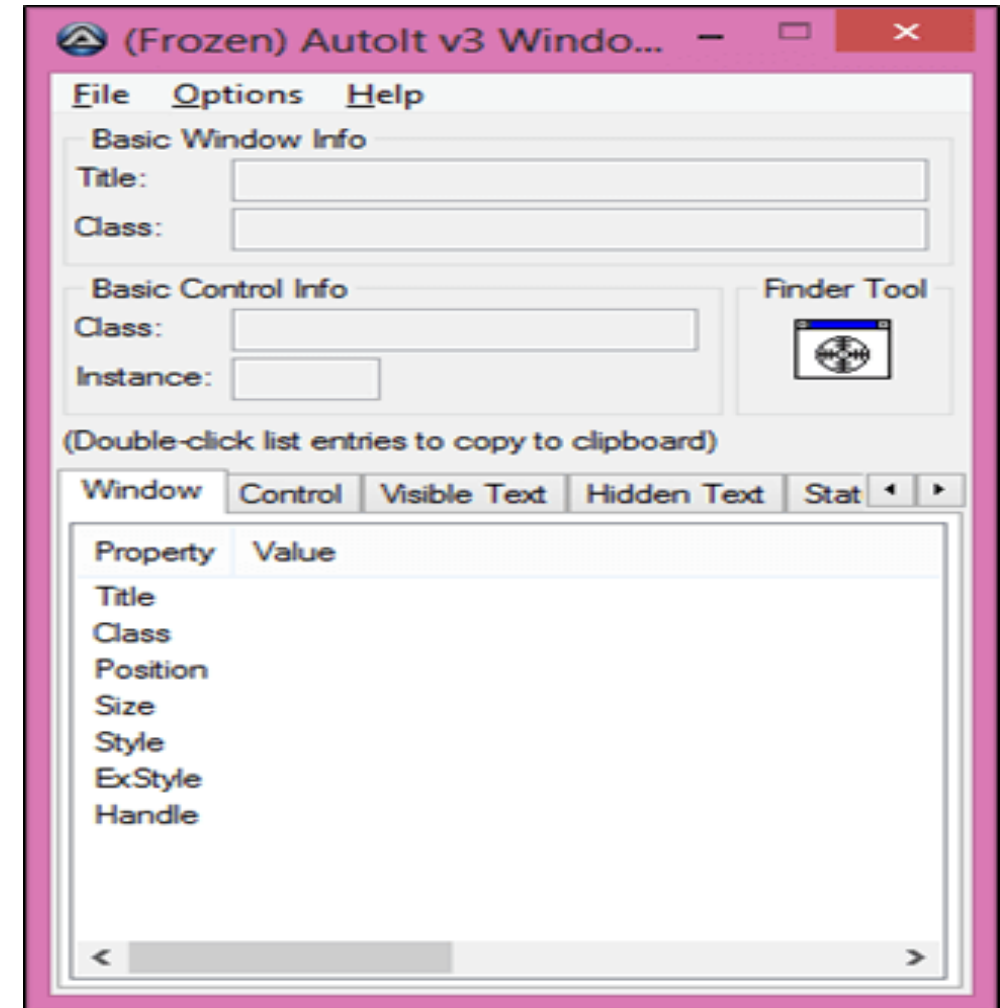
Find element using Element Identifier



Finding element using Element Identifier

Finding Element with Element Identifier

- Now open element Identifier- Go to 'C:\Program Files (x86)\Autolt3' and click on 'Au3Info.exe' file



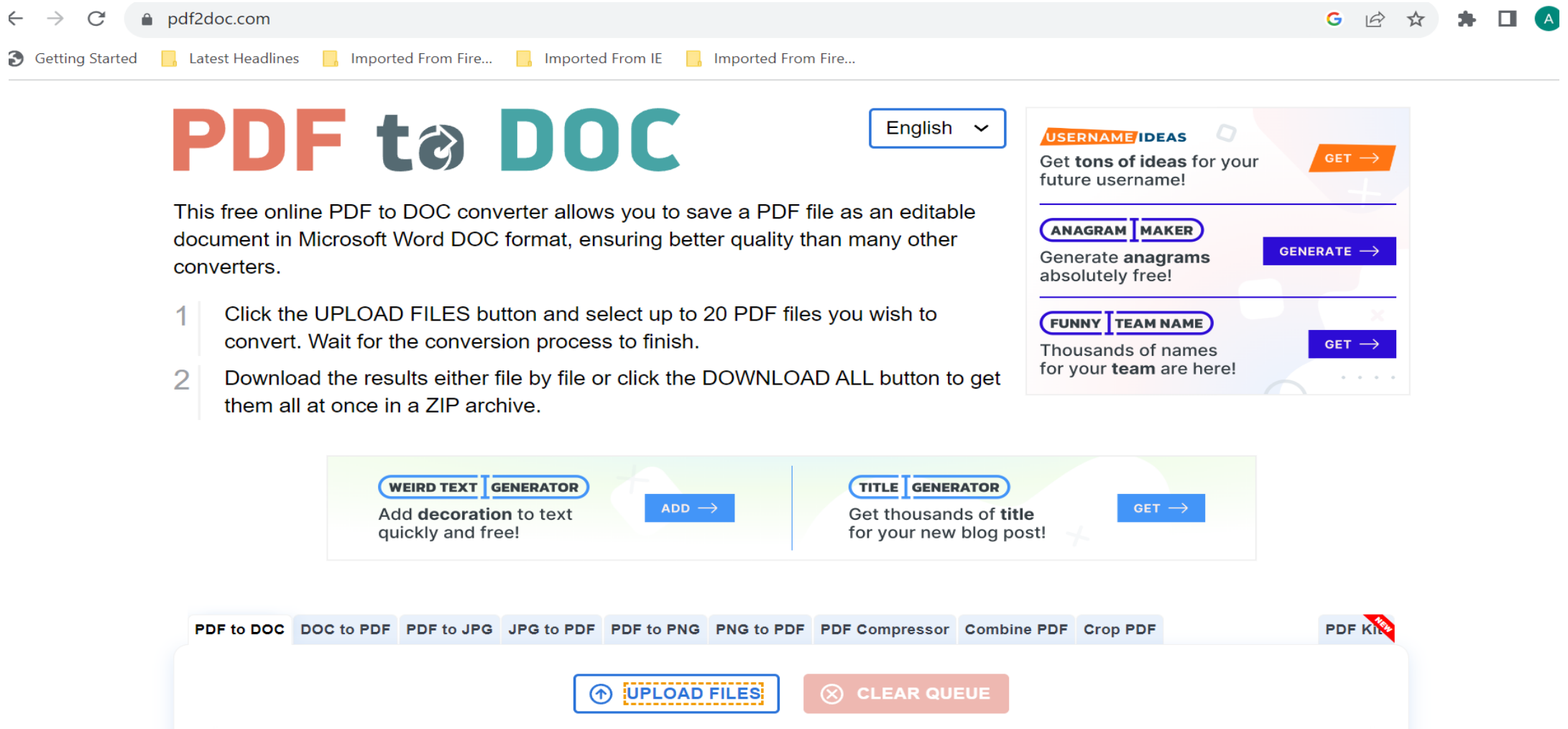
Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)

- Launch the following Web Application : <https://pdf2doc.com>
- Now open file uploader window by clicking on 'UPLOAD FILES' which is windows activity.

Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)



The screenshot shows the pdf2doc.com website. The main heading is "PDF to DOC" with a language dropdown set to "English". Below the heading, a description states: "This free online PDF to DOC converter allows you to save a PDF file as an editable document in Microsoft Word DOC format, ensuring better quality than many other converters."

Two numbered steps are listed:

- 1 Click the UPLOAD FILES button and select up to 20 PDF files you wish to convert. Wait for the conversion process to finish.
- 2 Download the results either file by file or click the DOWNLOAD ALL button to get them all at once in a ZIP archive.

On the right side, there are several tool cards:

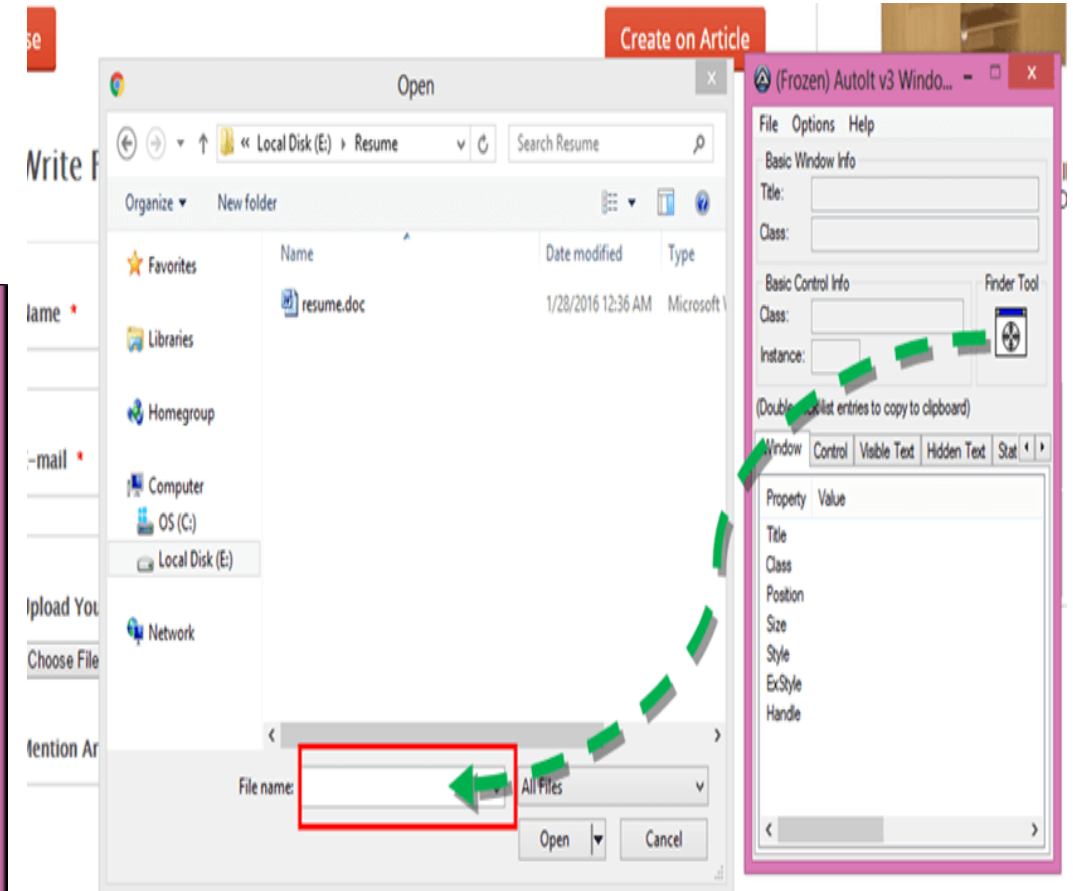
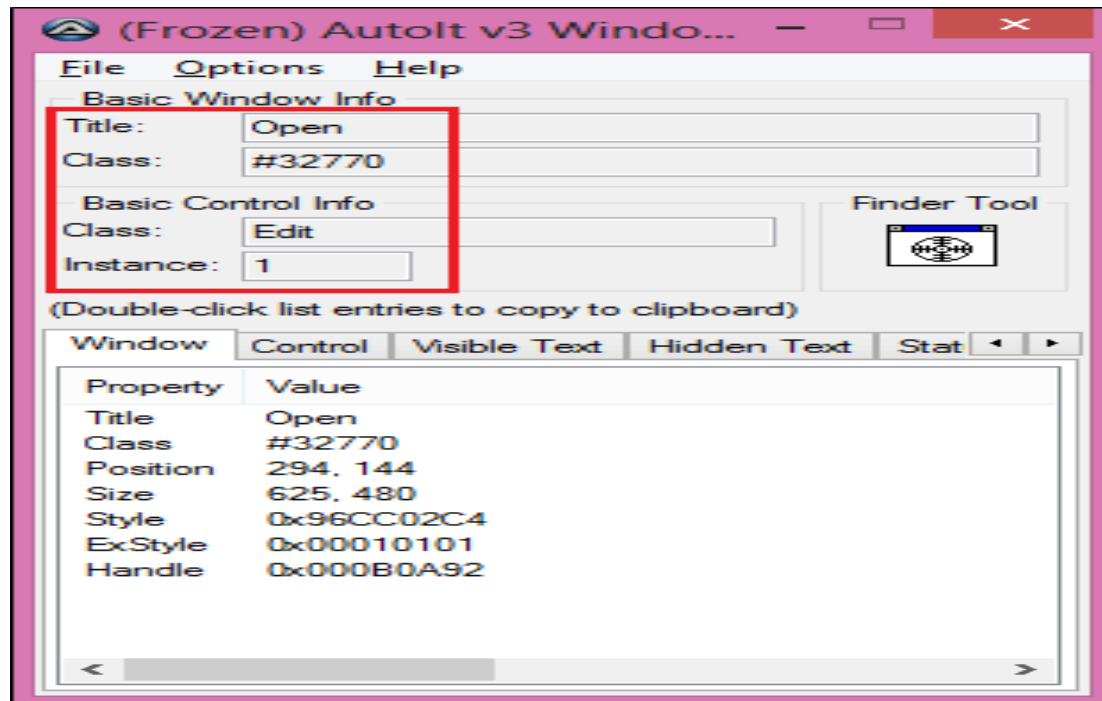
- USERNAME IDEAS**: Get tons of ideas for your future username! (GET →)
- ANAGRAM MAKER**: Generate anagrams absolutely free! (GENERATE →)
- FUNNY TEAM NAME**: Thousands of names for your team are here! (GET →)
- WEIRD TEXT GENERATOR**: Add decoration to text quickly and free! (ADD →)
- TITLE GENERATOR**: Get thousands of title for your new blog post! (GET →)

At the bottom, there is a navigation bar with links: PDF to DOC, DOC to PDF, PDF to JPG, JPG to PDF, PDF to PNG, PNG to PDF, PDF Compressor, Combine PDF, Crop PDF, and PDF KIL (with a "New" tag). Below the navigation bar are two buttons: "UPLOAD FILES" (with an upload icon) and "CLEAR QUEUE" (with a close icon).

Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)

- Drag the finder tool on the " File Name" box element of file uploader window to find the basic attributes.



Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)

Let's write scripts using AutoIT editor

Methods are required for writing file upload script:

Sets input focus to a given control on a window

ControlFocus(" title "," text ",controlID)

Sets text of a control

ControlSetText(" title "," text ",controlID ," File path which need to upload ")

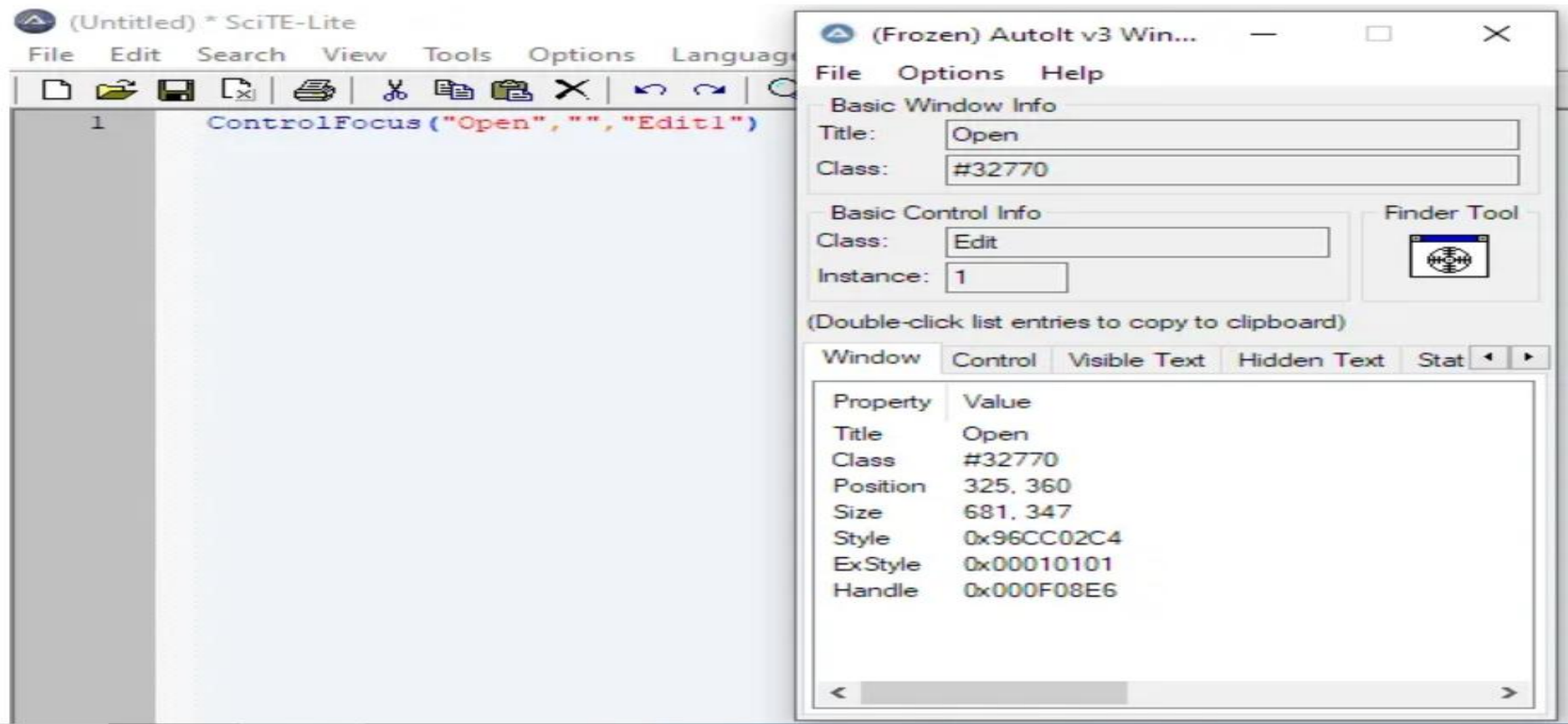
Sends a mouse click command to a given control

ControlClick(" title "," text ",controlID)

Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)

ControlFocus method: focus to the 'file name' text box of the file uploader



Finding Element with Element Identifier (Contd.)

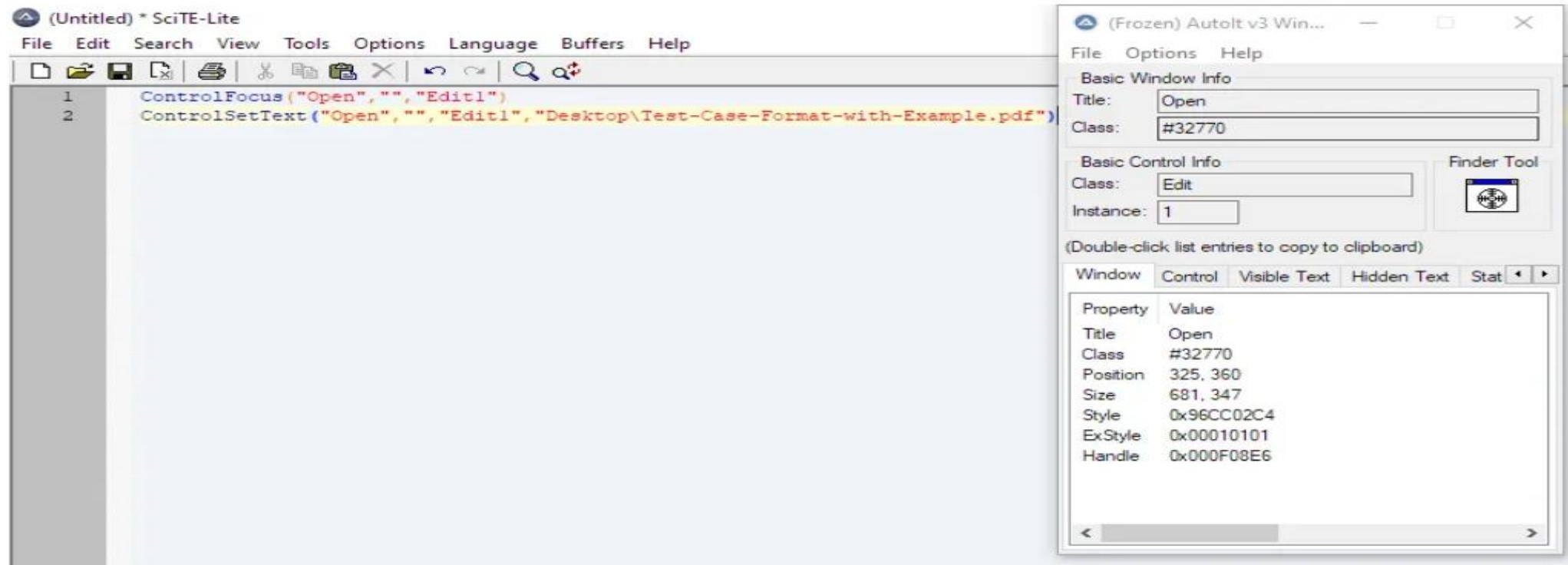
- **title** — “Open”
- **text** is not required.
- **controlID** — “Edit1” (combination of class='Edit' and Instance='1')

```
ControlFocus("Open","", "Edit1")
```

Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)

ControlSetText method :input file path of a control



ControlSetText method

Finding Element with Element Identifier (Contd.)

- **title** — “Open”
- **text** is not required.
- **controlID** — “Edit1”
- **new text** — pass the path of the file which we need to upload.

```
ControlSetText(“Open”,,””,”Edit1”,”Desktop\Test-Case-Format-with-Example.pdf”)
```

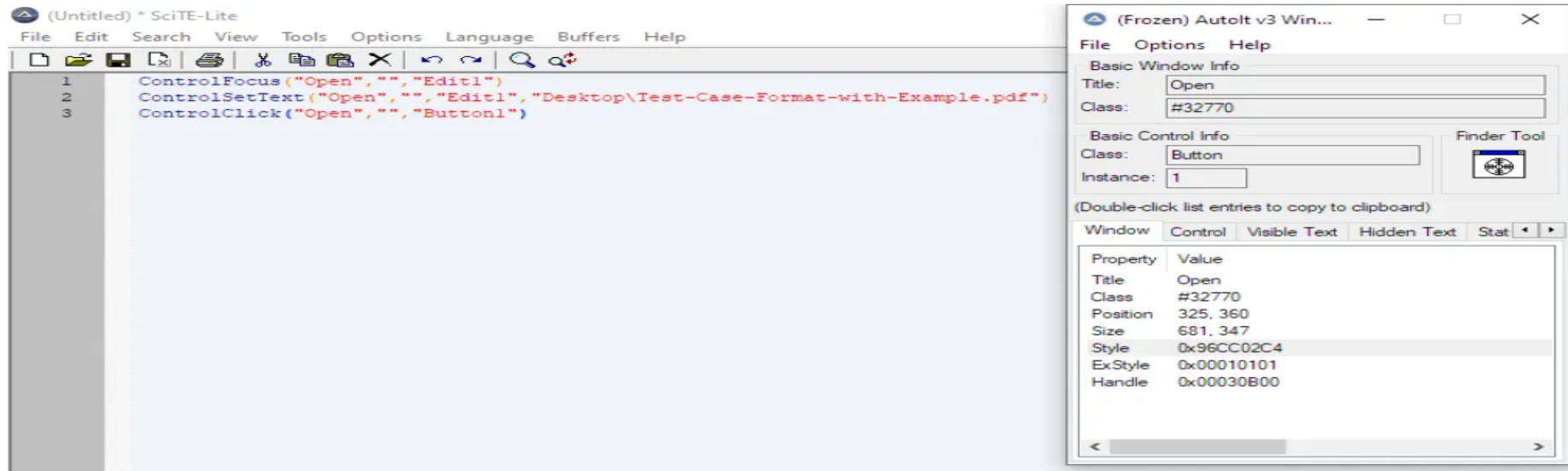
Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)

Step 4: Drag the finder tool on the “Open” button element of file uploader to find the attributes.

Attributes are **title='Open'**, **class='Button'** and **instance='1'**.

ControlClick method: click on ‘Open’ button of file uploader.



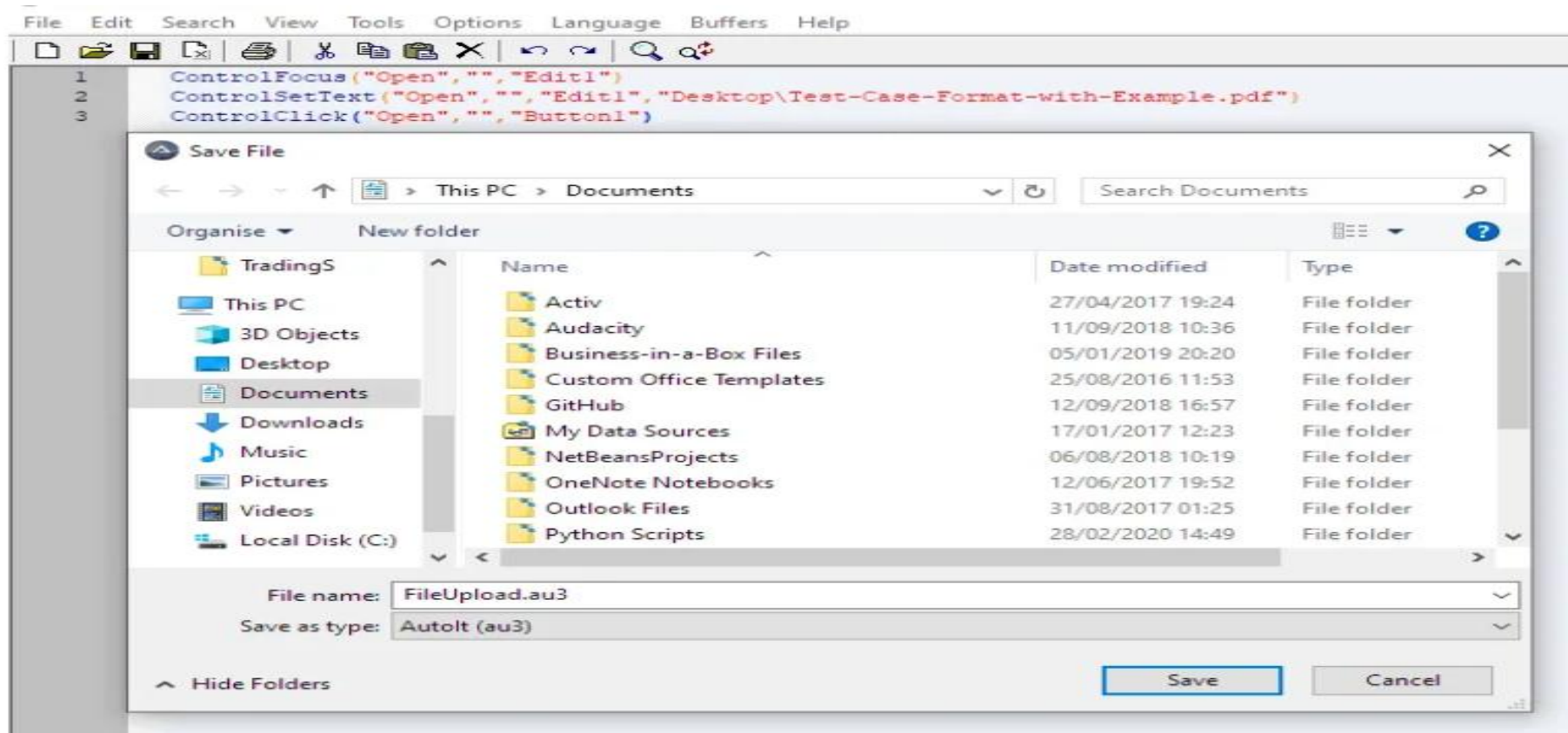
Finding Element with Element Identifier (Contd.)

- **title** — “ Open “
- **text** is not required.
- **controlID** — “ Button1 “ (combination of class and Instance)

```
ControlClick("Open","", "Button1")
```

Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)

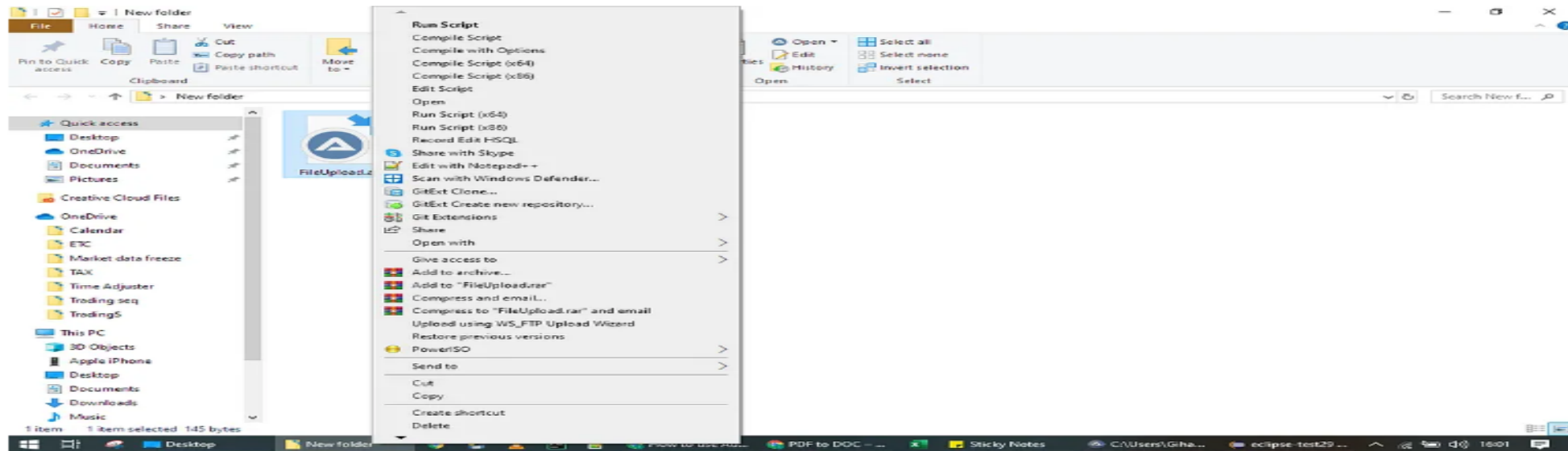


Save the script

Step 5: Save the script as “FileUpload”.

Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)



Compile the script

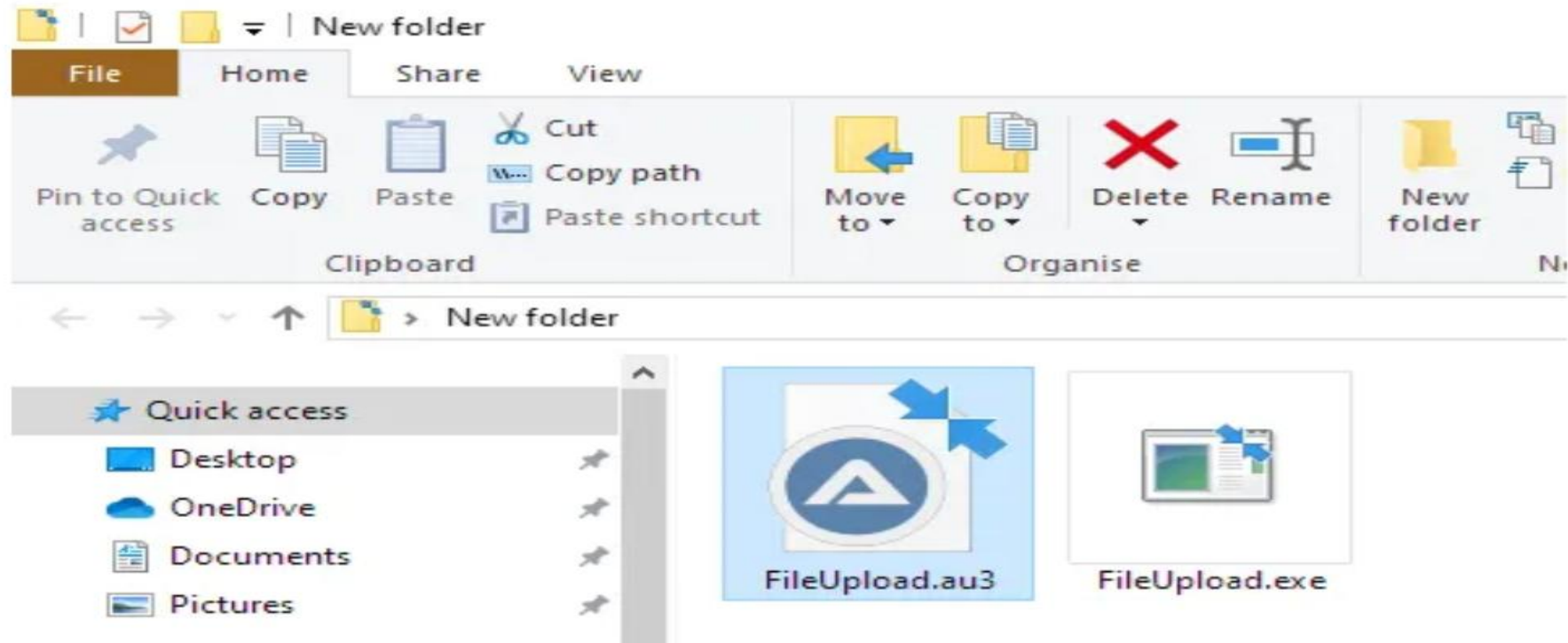
We can't execute this script directly, we need to compile this script. For compiling this script, you have two options:

1st Option : compile script x64 (for windows 64-bit machine)

2nd Option : compile script x86 (for windows 32-bit machine)

Finding element using Element Identifier

Finding Element with Element Identifier (Contd.)



'FileUpload exe' file generated after compilation

Step 6: 'FileUpload exe' file generated after compilation.

What is Auto IT?

AutoIT in Selenium

- In Selenium script, we find the elements of the application and upload file by executing AutoIT exe file generated from AutoIT script.
- When selenium clicks on 'UPLOAD FILES' button, file uploader box opens.
- Then we call AutoIT script, the control immediately transferred to AutoIT in order to upload a file and then control send back to selenium.

//call AutoIT script in selenium and upload file .

Runtime.getRuntime().exec("E:\\AutoIT\\FileUpload.exe");

- **Runtime** class allows the script to interface with the environment
- **getRuntime()** get the current runtime
- **exec()** methods execute the AutoIT script

What is Auto IT?

AutoIT in Selenium – Sample Program

```
import java.awt.AWTException;
import java.awt.Robot;
import java.awt.Toolkit;
import java.awt.datatransfer.StringSelection;
import java.awt.event.KeyEvent;
import java.io.IOException;
import java.time.Duration;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.edge.EdgeDriver;
import org.openqa.selenium.edge.EdgeOptions;
public class autoItDemo {
    public static void main(String[] args) throws InterruptedException, IOException,
    AWTException {
        EdgeOptions options = new EdgeOptions();
```

What is Auto IT?

AutoIT in Selenium

```
// ChromeOptions options = new ChromeOptions();
WebDriver driver = new EdgeDriver(options);
// WebDriver driver = new ChromeDriver(options);
driver.manage().window().maximize();
driver.manage().timeouts().implicitlyWait(Duration.ofSeconds(10));
driver.get("https://www.pdf2doc.com");
//Thread.sleep(5000);
WebElement file = driver.findElement(By.xpath("//span[text()='UPLOAD FILES']"));
file.click();
//Method1 using AutoIT
// Thread.sleep(5000);
Runtime.getRuntime().exec("C:/arun/Newfolder/AutoItFileUpload1.exe");

// driver.close();
```

What is Auto IT?

AutoIT in Selenium

```
// creating object of Robot class
//Method2 using Robot Class
Robot rb = new Robot();
// copying File path to Clipboard
StringSelection str = new StringSelection("file:///C:/arun/Newfolder/Test.pdf");
Toolkit.getDefaultToolkit().getSystemClipboard().setContents(str, null);
// press Contol+V for pasting
rb.keyPress(KeyEvent.VK_CONTROL);
rb.keyPress(KeyEvent.VK_V);
// release Contol+V for pasting
rb.keyRelease(KeyEvent.VK_CONTROL);
rb.keyRelease(KeyEvent.VK_V);
// for pressing and releasing Enter
rb.keyPress(KeyEvent.VK_ENTER);
rb.keyRelease(KeyEvent.VK_ENTER);
}
}
```

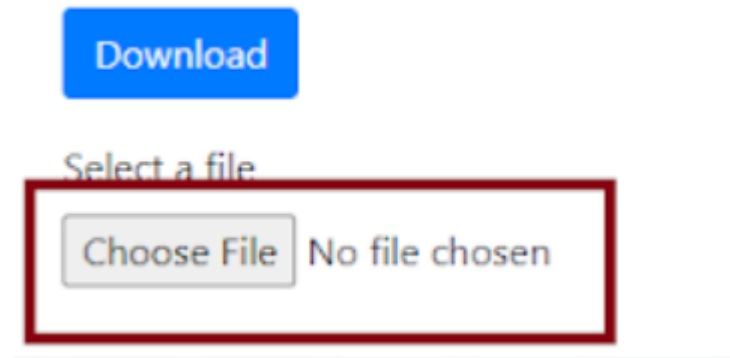
File Upload in Selenium



File Upload

File Upload

- We often face the problem where we have to upload a file in Web Application.
- This can be easily achieved in Selenium. We are using sendKeys() method to set the value of the file to upload.

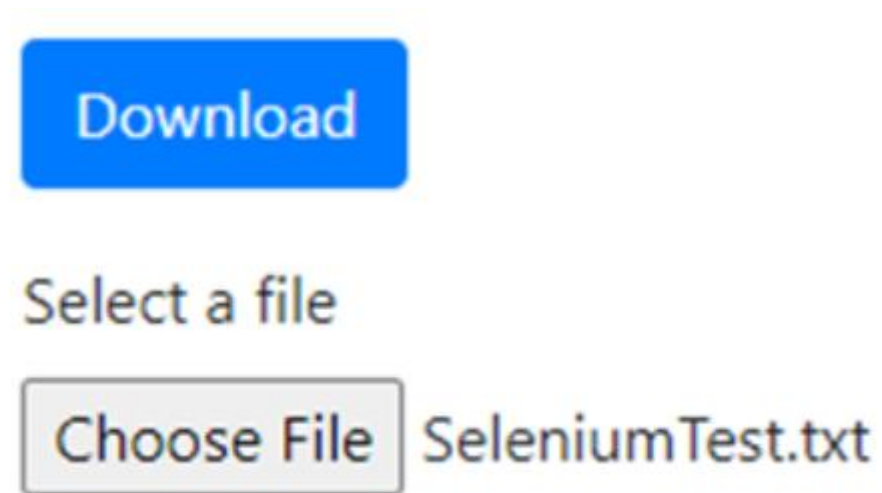


File Upload in Selenium – Sample Program

```
public class Upload_Demo {  
    public static void main(String[] args) {  
  
        ChromeOptions options = new ChromeOptions();  
        WebDriver driver = new ChromeDriver(options);  
        driver.manage().window().maximize();  
        driver.get("https://demoqa.com/upload-download");  
  
        // Locating upload button  
        WebElement upload = driver.findElement(By.id("uploadFile"));  
  
        //Upload the file  
        upload.sendKeys("C:\\Documents\\SeleniumTest.txt");  
  
        String Message = driver.findElement(By.id("uploadedFilePath")).getText();  
        System.out.println("Message is :" + Message);  
        driver.close();  
    }  
}
```

File Upload

File Upload in Selenium – Output



PDF Download in Chrome with Selenium



PDF Download

- When a PDF link on a website is clicked, the PDF file typically opens in Chrome's default PDF viewer.
- While clicking on the PDF link itself is straightforward using Selenium, interacting with the PDF viewer afterward presents a challenge.
- This is because the PDF viewer is not part of the HTML DOM, and Selenium cannot interact with elements within it, such as the download button.
- As a result, we are unable to perform Selenium actions on controls within Chrome's built-in PDF viewer.
- To overcome this problem, we need to disable the default chrome pdf viewer plugin before launching the driver, and set the download directory where we need to download the file.

PDF Download in Selenium – Sample Program

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;

import java.io.File;
import java.util.HashMap;
import java.util.Map;

public class DownloadPDF_Chrome_Demo {

    public static void main(String[] args) throws InterruptedException {

        String downloadFilepath = System.getProperty("user.dir") + File.separator
+ "downloads";
```

PDF Download in Selenium – Sample Program

```
ChromeOptions options = new ChromeOptions();
Map<String, Object> prefs = new HashMap<>();
prefs.put("plugins.always_open_pdf_externally", true);
prefs.put("download.default_directory", downloadFilepath);
options.setExperimentalOption("prefs", prefs);

WebDriver driver = new ChromeDriver(options);
driver.manage().window().maximize();
driver.get("https://freetestdata.com/document-files/pdf/");

// Locate and click the download link or button if necessary
WebElement downloadLink =
driver.findElement(By.xpath("//*[@class=\"elementor-button-text\"]"));
downloadLink.click();
```

PDF Download in Selenium – Sample Program

```
        // Wait for the download to complete
        Thread.sleep(5000);

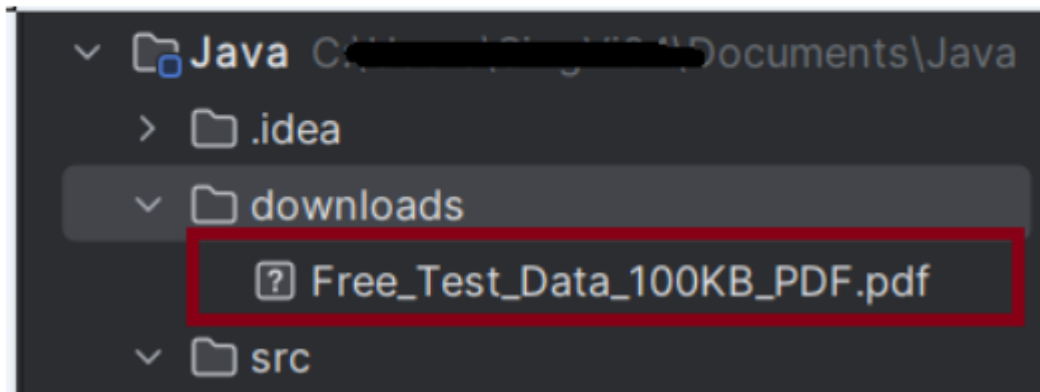
        // Check if the file exists
        File downloadedFile = new File(downloadFilepath +
"/Free_Test_Data_100KB_PDF.pdf");
        if (downloadedFile.exists()) {
            System.out.println("File is downloaded!");
        } else {
            System.out.println("File is not downloaded.");
        }

        driver.quit();
    }
}
```

PDF Download in Selenium – Output

```
File is downloaded!  
  
Process finished with exit code 0
```

We can see that a downloads folder is created, and it has the downloaded pdf file – ***Free_Test_Data_100KB_PDF.pdf***.



Read PDF in Chrome with Selenium



Read PDF

- Organizations frequently generate various types of PDF reports, such as mobile bills, electricity bills, financial reports, and revenue reports.
- Quality Assurance (QA) teams are then tasked with verifying the information contained in these reports.
- Typically, this process involves manually downloading the reports and reading the data they contain. To automate this process, the test framework must be capable of automatically downloading PDF reports and extracting the data without any human intervention.
- Reading a PDF document in Selenium using Java requires some additional libraries because Selenium itself does not provide direct support for reading PDFs.
- The most commonly used library for reading PDFs in Java is Apache PDFBox.

Read PDF – Dependency to be added in POM.xml

```
<dependency>  
    <groupId>org.seleniumhq.selenium</groupId>  
    <artifactId>selenium-java</artifactId>  
    <version>4.24.0</version>  
</dependency>
```

```
<dependency>  
    <groupId>commons-io</groupId>  
    <artifactId>commons-io</artifactId>  
    <version>2.16.1</version>  
</dependency>
```

```
<dependency>  
    <groupId>org.apache.pdfbox</groupId>  
    <artifactId>pdfbox</artifactId>  
    <version>3.0.3</version>  
</dependency>
```

Read PDF in Selenium – Sample Program

```
import org.apache.pdfbox.Loader;  
import org.apache.pdfbox.pdmodel.PDDocument;  
import org.apache.pdfbox.text.PDFTextStripper;  
import org.openqa.selenium.By;  
import org.openqa.selenium.WebDriver;  
import org.openqa.selenium.WebElement;  
import org.openqa.selenium.chrome.ChromeDriver;  
import org.openqa.selenium.chrome.ChromeOptions;  
import org.openqa.selenium.support.ui.ExpectedCondition;  
import org.openqa.selenium.support.ui.WebDriverWait;  
  
import java.io.File;  
import java.io.IOException;  
import java.time.Duration;  
import java.util.HashMap;  
import java.util.Map;
```

Read PDF in Selenium – Sample Program

```
public class ReadPDF_Chrome_Demo {  
  
    public static void main(String[] args) throws InterruptedException {  
  
        String downloadFilepath = System.getProperty("user.dir") + File.separator  
        + "chrome_downloads";  
  
        ChromeOptions options = new ChromeOptions();  
        Map<String, Object> prefs = new HashMap<>();  
        prefs.put("plugins.always_open_pdf_externally", true);  
        prefs.put("download.default_directory", downloadFilepath);  
        options.setExperimentalOption("prefs", prefs);  
  
        WebDriver driver = new ChromeDriver(options);  
        driver.manage().window().maximize();  
        driver.get("https://freetestdata.com/document-files/pdf/");  
    }  
}
```

Read PDF in Selenium – Sample Program

```
WebElement downloadLink = driver.findElement(By.xpath("//*[@class='elementor-button-text']"));
downloadLink.click();

//Wait for download to complete
File downloadedFile = new File(downloadFilepath +
                                "/Free_Test_Data_100KB_PDF.pdf");
WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(30));
wait.until((ExpectedCondition<Boolean>) wd -> downloadedFile.exists());

// Check if the file exists
if (downloadedFile.exists()) {
    System.out.println("File is downloaded!");
} else {
    System.out.println("File is not downloaded.");
}

driver.quit();
```

Read PDF in Selenium – Sample Program

```
// Read the downloaded PDF using PDFBox
PDDocument document = null;
try {
    document = Loader.loadPDF(downloadedFile);
    PDFTextStripper pdfStripper = new PDFTextStripper();
    String text = pdfStripper.getText(document);
    document.close();
    // Print the PDF text content
    System.out.println("Text in PDF: ");
    System.out.println(text);
} catch (IOException e) {
    System.err.println("An error occurred while loading or reading the PDF
    file: " + e.getMessage());
    e.printStackTrace();
}
}
```

PDF Download in Selenium – Output

```
ReadPDF_Chrome_Demo x
File is downloaded!
Text in PDF:
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Donec iaculis elit nec ante eleifend,
eget cursus tortor auctor. Integer at ultrices lorem, eget bibendum turpis. Praesent lacus
sapien, ullamcorper gravida suscipit eu, pharetra et eros. Integer interdum dictum volutpat.
Nulla pellentesque ante vitae sapien luctus ornare. Ut a risus quis diam rhoncus rhoncus.
Integer tortor lorem, vulputate non tempus eu, suscipit id tellus. In laoreet sollicitudin quam, et
condimentum justo ultrices eget. Pellentesque hendrerit suscipit diam ac porttitor. Aenean
dictum id magna vel venenatis. In rhoncus tempus libero, sed consectetur nibh ultrices nec.
Aenean pulvinar euismod ligula at lacinia. Ut consectetur dui ipsum, a rhoncus lacus gravida
vitae. In mollis tortor in libero lobortis molestie. Integer tempor justo ligula, eu euismod lectus
```

THANK YOU